

Fusion Compiler Project

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Gaëtan Jenni (386450)

Q1

What is the critical path in each compilation step ?

1. Initial map: op_a_i[3] → result_q_reg[31] (0.2787 ns)
2. Logic opto: op_a_i[8] → result_q_reg[31] (0.2750 ns)
3. Final opto interconnect: op_a_i[8] → result_q_reg[31] (0.4669 ns)
4. Final opto: op_a_i[22] → result_q_reg[31] (0.4282 ns)

Write the chain of cells of the critical path

```
1 op_a_i[22] -> A8532/X -> ctm_i_7060/X -> ctm_i_7059/X -> A7995/X -> DP_OP_4_61613_15352_J1/U_1350/C0 ->
DP_OP_4_61613_15352_J1/U_1368/S -> DP_OP_4_61613_15352_J1/U_1372/C0 -> ctmTdsLR_2_11147/X -> ctmTdsLR_2_10048/X ->
ctmTdsLR_1_10158/X -> ctmTdsLR_3_10160/X -> ctmTdsLR_4_10356/X -> result_q_reg[31]/D
```

How long it is ?

There are 12 cells between the start and the destination (so 14 if we count them).
The critical path in time is 0.4282 ns.

What is the utilization ?

Utilization Ratio: 0.8743, so 87.43%.

Q2

What is the critical path in each compilation step ?

The script reports only the "final opto":

- Final opto: op_a_i[12] → result_q_reg[31] (0.4257 ns)

Write the chain of cells of the critical path

```
1 op_a_i[12] -> ctmTdsLR_2_5380/X -> ZINV_644_inst_6016/X -> mult_18/ctmi_4038/X -> mult_18/A5053/X -> mult_18/U_1340/S ->
mult_18/U_1344/S -> ctmTdsLR_13_7385/X -> ctmTdsLR_19_7391/X -> mult_18/U_1360/S -> mult_18/ctmi_3236/X ->
ctmTdsLR_1_5986/X -> ctmTdsLR_2_5987/X -> ctmTdsLR_1_5831/X -> mult_18/A5314/X -> ctmTdsLR_3_5456/X ->
ctmTdsLR_4_5457/X -> mult_18/A5315/X -> result_q_reg[31]/D
```

How long it is ?

There are 17 cells between the start and the destination (so 19 if we count them).
The critical path in time is 0.4257 ns.

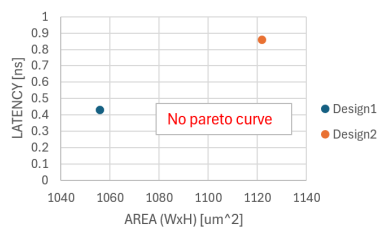
What is the utilization ?

Utilization Ratio: 0.7721, so 77.21%.

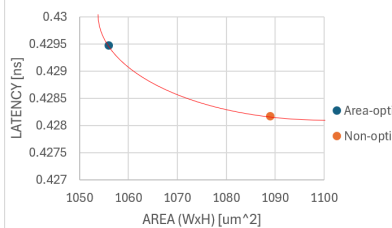
Q3-4

DESIGN	AREA(WxH)[μm^2]	LATENCY[ns]
design1, nonOptimized(fromQ1)	1089	0.4282
design2, nonOptimized(fromQ2)	1296	0.8513
design1, areaOptimized	1056	0.4295
design2, areaOptimized	1122	0.8589

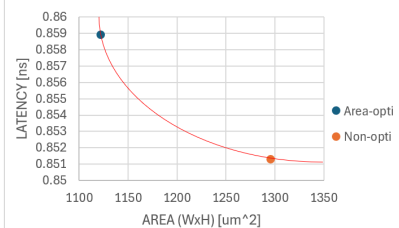
Latency vs Area for design 1 & 2 area-opti



Latency vs Area for design 1



Latency vs Area for design 2



Q5

What is the critical path in each compilation step ?

- Final opto: op_a_i[3] → result_q_reg[31] (1.3787 ns)

Write the chain of cells of the critical path

```

1 op_a_i[3] -> HFSINV_3815_391/X -> ctmi_176/X -> ctmi_174/X -> ctmi_177/X -> DP_OP_4_61613_15352_J1/U_1140/S ->
DP_OP_4_61613_15352_J1/U_1141/S -> DP_OP_4_61613_15352_J1/U_1402/CO -> DP_OP_4_61613_15352_J1/U_1405/CO ->
DP_OP_4_61613_15352_J1/U_1407/CO -> DP_OP_4_61613_15352_J1/U_1410/CO -> DP_OP_4_61613_15352_J1/U_1412/CO ->
DP_OP_4_61613_15352_J1/U_1415/CO -> DP_OP_4_61613_15352_J1/U_1417/CO -> DP_OP_4_61613_15352_J1/U_1420/CO ->
DP_OP_4_61613_15352_J1/U_1422/CO -> DP_OP_4_61613_15352_J1/U_1425/CO -> DP_OP_4_61613_15352_J1/U_1427/CO ->
DP_OP_4_61613_15352_J1/U_1430/CO -> DP_OP_4_61613_15352_J1/U_1432/CO -> DP_OP_4_61613_15352_J1/U_1435/CO ->
DP_OP_4_61613_15352_J1/U_1437/CO -> DP_OP_4_61613_15352_J1/U_1440/CO -> DP_OP_4_61613_15352_J1/U_1442/CO ->
DP_OP_4_61613_15352_J1/U_1445/CO -> DP_OP_4_61613_15352_J1/U_1447/CO -> DP_OP_4_61613_15352_J1/U_1450/CO ->
DP_OP_4_61613_15352_J1/U_1452/CO -> DP_OP_4_61613_15352_J1/U_1455/CO -> DP_OP_4_61613_15352_J1/U_1457/CO ->
DP_OP_4_61613_15352_J1/U_1460/CO -> DP_OP_4_61613_15352_J1/U_1462/CO -> DP_OP_4_61613_15352_J1/U_1465/CO ->
DP_OP_4_61613_15352_J1/U_1467/CO -> DP_OP_4_61613_15352_J1/ctmi_2477/X -> DP_OP_4_61613_15352_J1/ctmi_2476/X ->
ctmi_178/X -> result_q_reg[31]/D

```

How long it is ?

There are 36 cells between the start and the destination (so 38 if we count them).
The critical path in time is 1.3787 ns.

What is the utilization ?

Utilization Ratio: 0.8692, so 86.92% (with a floorplan of 25x25).

What is the throughput ?

The throughput is 1.

What is the latency ?

The latency is 1 cycle, so it's equal to the critical path which is 1.3787 ns.

Comparison with previous design

The latency has increased significantly from 0.42 ns to 1.3787 ns, but the area has decreased to 952 μm^2 .

Q6

What is the critical path in each compilation step ?

- Final opto: op_a_i[14] $\rightarrow$ mac_res_q_reg[31] (0.4216 ns)

Write the chain of cells of the critical path

```

1 op_a_i[14] -> ctmTdsLR_1_5866/X -> ctmi_3697/X -> ctmTdsLR_1_5757/X ->
ctmTdsLR_2_4857/X -> DP_OP_20_58942_35926_J1/ctmi_3584/X ->
DP_OP_20_58942_35926_J1/U_416/CO -> DP_OP_20_58942_35926_J1/U_427/S ->
DP_OP_20_58942_35926_J1/U_429/CO -> DP_OP_20_58942_35926_J1/U_438/S ->
DP_OP_20_58942_35926_J1/ctmi_3329/X -> DP_OP_20_58942_35926_J1/A4770/X ->
ctmTdsLR_2_4969/X -> ctmTdsLR_3_4970/X -> ctmTdsLR_3_4902/X ->
DP_OP_20_58942_35926_J1/A4189/X -> ctmTdsLR_1_5120/X ->
ctmTdsLR_6_5024/X -> mac_res_q_reg[31]/D

```

How long it is ?

There are 17 cells between the start and the destination (so 19 if we count them).
The critical path in time is 0.4216 ns.

What is the utilization ?

Utilization Ratio: 0.9087, so 90.87% (with a floorplan of 31x31).

What is the throughput ?

The throughput is 1/3.

What is the latency ?

Latency = 3 cycles = $3 * 0.4216 \text{ ns} = 1.2648 \text{ ns}$

Explain the different architectures with PROs and CONS

The **mac_unit_1** compute the full 32×32 multiplication and addition in one cycle, giving the lowest latency but the longest critical path.

The **mac_unit_2** splits the multiplication and addition into two clock cycles, but it doesn't reduce significantly the critical path compared to **mac_unit_1**, and it adds also more area due to the extra registers.

The **mac_unit_3** uses a multi-cycle FSM that computes the 32×32 product from smaller 16×16 partial products, reducing area and also a bit the critical path. However, it increases the latency (3 cycles).

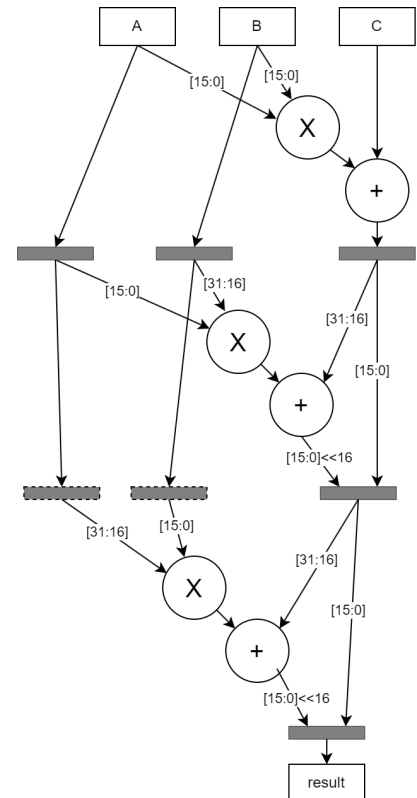


Figure 1: Dataflow graph

Appendices

The following script was created to automatize the floorplan optimization. It tries different candidate floorplans from the largest to the smallest until it finds the smallest one that still allows the compilation to succeed.

```

1  ##### Sourcing common setup script
2  source -echo ../scripts/project_fc_common_setup.tcl
3
4  ##### Sourcing flow setup script
5  source -echo ../../setup/fc_flow_setup.tcl
6
7  ##### Define global variables
8  set EXPORT_NAME "project6_mac_unit_3"
9  set DESIGN_NAME "mac_unit_3"
10 set SV_FILE "mac_unit_3.sv"
11 set SDC_FILE_MAC_UNIT_NAME "mac_unit.sdc"
12 set CREATE_BLOCK false
13
14 # Open the design library
15 open_lib ${RESULTS_PATH}/${DESIGN_LIBRARY}
16
17 ##### Report reference libraries
18 report_ref_libs
19
20 if ${CREATE_BLOCK} == true {
21     ##### Reading RTL
22
23     # Suppress known warnings
24     suppress_message VER-130
25
26     analyze -format sverilog ${SV_FILE}
27
28     # Unsuppress after analyze stage
29     unsuppress_message VER-130
30
31     # Elaborate
32     elaborate ${DESIGN_NAME}
33
34     # Set top module in the design
35     set_top_module ${DESIGN_NAME}
36
37     save_block -as ${DESIGN_NAME}/${EXPORT_NAME}_read_rtl
38     close_block -force
39 }
40
41 # Define candidate shapes and sizes
42 # Must be tried in order from largest to smallest
43 set candidate_floorplans {
44     {R {33 33} }
45     {R {32 33} }
46     {R {32 32} }
47     {R {31 32} }
48     {R {31 31} }
49     {R {30 31} }
50     {R {30 30} }
51 }
52
53 set last_successful ""
54
55 foreach fp $candidate_floorplans {
56     set shape [lindex $fp 0]
57     set size_list [lindex $fp 1]
58     set width [lindex $size_list 0]
59     set height [lindex $size_list 1]
60
61     puts "Trying floorplan shape ${shape}, size ${width} x ${height}"
62
63     open_block ${DESIGN_NAME}/${EXPORT_NAME}_read_rtl
64
65     # Source tech setup script
66     source -echo ../../setup/tech_setup.tcl
67
68     # Setup application options
69     set_lib_cell_purpose -include none {*/*_A021* */*_V2LP*}
70     set_app_options -name place.coarse.continue_on_missing_scandef -value true
71     set_app_options -name compile.flow.enable_ccd -value false
72
73     # Initialize floorplan
74     initialize_floorplan -control_type core -core_utilization 0.6 -core_offset 5 -shape $shape -side_length $size_list
75     -flip_first_row true
76
77     set_block_pin_constraints -self -allowed_layers {M4 M3} -sides {1 4} -pin_spacing_distance 1 -width 0.11 -length 0.11
78
79     set_individual_pin_constraints -ports [get_ports clk_i] -sides 2 -allowed_layers M5
80
81     # Create power/ground ports only if they do not already exist
82     # Using llength on get_ports will return 0 if not present
83     if {[sizeof_collection [get_ports VDD]] == 0} {
84         create_port -port_type power -direction inout VDD
85         puts "Created power port VDD"
86     } else {
87         puts "Port VDD already exists - skipping create_port."
88     }
89
90     if {[sizeof_collection [get_ports VSS]] == 0} {
91         create_port -port_type ground -direction inout VSS
92         puts "Created ground port VSS"
93     } else {
94         puts "Port VSS already exists - skipping create_port."
95     }
96
97     # Create power/ground nets if they don't exist
98     # create_net returns error if net exists; protect with catch and existence check

```

```

98  if { [sizeof_collection [get_nets VDD]] == 0 } {
99      create_net -power VDD
100      puts "Created power net VDD"
101  } else {
102      puts "Net VDD already exists - skipping create_net."
103  }
104
105  if { [sizeof_collection [get_nets VSS]] == 0 } {
106      create_net -ground VSS
107      puts "Created ground net VSS"
108  } else {
109      puts "Net VSS already exists - skipping create_net."
110  }
111
112  ##### Set pin placement constraints
113  set ports [remove_from_collection [get_ports] {VDD VSS}]
114
115  place_pins -self -ports ${ports}
116
117  ##### Insert Boundary/TAP cells in the design
118  source -echo ../scripts/insert_boundary_and_tap_cells.tcl
119
120  ##### Create Power/Ground Network
121  source -echo ../scripts/create_pg_network.tcl
122
123  # MCM setup
124  set SDC_FILE_MAC_UNIT "${SDC_FILE_MAC_UNIT_NAME}"
125  source -echo ../scripts/project_mcm_setup.tcl
126
127  set_app_options -name time.delay_calculation_style -value auto
128  set status [catch {compile_fusion -to final_opto} result]
129
130  if {$status == 1} {
131      puts "-----"
132      puts "Floorplan ${shape} ${width}x${height} FAILED:"
133      puts "Result: ${result}, status: ${status}"
134      puts "Smallest SUCCESSFUL floorplan so far: ${last_successful}"
135      puts "-----"
136      exit
137  } else {
138      puts "-----"
139      puts "Floorplan ${shape} ${width}x${height} SUCCESS"
140      puts "Result: ${result}, status: ${status}"
141      puts "-----"
142      set last_successful "${shape} ${width}x${height}"
143
144      # Reports
145      #for the final reports, we want to use zero_interconnect to be consistent with previous runs
146      set_app_options -name time.delay_calculation_style -value zero_interconnect
147
148      report_area > reports/area_final_opto_report_${EXPORT_NAME}.log
149      report_timing -significant_digits 10 > reports/timing_final_opto_report_${EXPORT_NAME}.log
150
151      report_resources > reports/resource_final_opto_report_${EXPORT_NAME}.log
152      report_utilization > reports/utilization_final_opto_report_${EXPORT_NAME}.log
153
154      save_block -as ${DESIGN_NAME}/${EXPORT_NAME}_final_opto
155  }
156 }
157
158 puts "-----"
159 puts "Smallest SUCCESSFUL floorplan so far: ${last_successful}"
160 puts "-----"
161
162 exit

```

Listing 1: Tcl script used for Q6